Acute Graft-Versus-Host Disease

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43.1 Introduction

Graft-versus-host disease (GvHD) was first recognized in murine models of HCT, and in the absence of knowledge of the HLA system, it was termed "secondary" (secondary to recovery from irradiation damage) or "runt" disease. Billingham established the criteria for the occurrence of secondary disease in the 1960s, i.e.:

- The administration of a graft containing immunocompetent cells
- Immunological disparity between host and donor
- The administration of the graft to an immunosuppressed host unable to reject the graft cells

In the human setting, we traditionally recognize two forms of GvHD, acute (aGvHD) and chronic (cGvHD). The original distinction of acute from chronic GvHD, namely, the occurrence before or after day 100 post stem cell infusion, has become blurred due to occurrence of aGvHD symptoms beyond day 100 after RIC regimens and/or after DLI (usually given after day 100). Nevertheless, the underlying combination of symptoms and signs affecting the skin, liver, and gastrointestinal tract forms a classical clinical syndrome enabling the diagnosis, and a helpful guide to the appropriate terminology is provided in Table 43.1 (Filipovich et al. 2005).

Table 43.1 Current classification of acute and chronic GvHD

Classification	Day after HCT	Features of acute GvHD	Features of chronic GvHD
Acute GvHD			
– Classic acute	<100 days	Yes	No
– Persistent, recurrent, or late onset	>100 days	Yes	No
Chronic GvHD			
– Classic chronic	No time limit	No	Yes
– Overlap syndrome	No time limit	Yes	Yes

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43.2 Definition

aGvHD remains, directly or indirectly, the major cause of short-term (day 100 and 1 year) mortality after allo-HCT. The pathophysiology of aGvHD has been attributed to a three-phase process comprising initial tissue damage from the conditioning regimen which in turn leads to activation of host antigen-presenting cells by pathogen-associated molecular patterns (PAMPs) and damage-associated molecular patterns (DAMPs) and activation and proliferation of donor T cells (afferent phase) and finally to the effector phase characterized by cytotoxic cell damage and release of inflammatory cytokines such as interleukin-1 (IL-1) and tissue necrosis factor-alpha (TNFα) that eventually produce tissue necrosis (efferent phase). The action of this pathogenetic process in the induction of aGvHD is modulated in part by the presence of cells capable of inhibiting immune response, such as T-regulatory cells (Tregs), type 1 regulatory T cells (Tr1 cells), invariant NKT cells, and myeloid-derived suppressor cells (MDSCs) (Ferrara et al. 2009; Teshima et al. 2016).

43.3 Risk Factors

As aGvHD is a result of an alloimmune effect, the major risk for occurrence is the presence of HLA disparity and increasing degrees of HLA mismatching increase the probability of more severe disease. Other important and consistent risk factors include older patient age, the use of female donors for male recipients, prior alloimmunization of the donor, and the nature of GvHD prophylaxis. A number of publications have variously reported risk factors such as increasing donor age, increasing intensity of the preparative regimen, the use of PBSC as opposed to BM, and recipient seropositivity for CMV.

A recent study of 2941 recipients of allo-HCT in Seattle confirmed the importance of the degree of HLA mismatching, the use of URD, and the administration of high-dose TBI in predicting the occurrence of moderate to severe aGvHD. In contrast they found that increasing donor age, cytokine-mobilized stem cells, and the use of female donors for male recipients did not impact on the likelihood of aGvHD but were associated with the occurrence of cGvHD (Flowers et al. 2011).

More recently, we have begun to appreciate the importance of non-HLA genetic factors in the development of GvHD. Examples include polymorphisms in the genes encoding cytokines such as the tumor necrosis factors, the interleukins (IL-1, IL-6, and IL-10), interferon gamma (IFN-γ), and transforming growth factor-β3 (TGF-β3) and the expression of the killer cell immunoglobulin-like receptors (KIR). Interestingly, one of the common features of the organs involved in aGvHD is that they are all exposed to microbial pathogens through the intestinal mucosa, epidermis, and portal circulation, and early murine studies confirmed a reduction in the severity and incidence of GvHD in animals that received antibiotic prophylaxis to "decontaminate" the GI tract or those kept in germ-free environments. This has led to the speculation that potential differences within individuals in the interactions of antigens derived from infective organisms and pathogen recognition receptors (PRR) might protect or predispose to the occurrence of GvHD. This concept was further supported by studies on SNPs within NOD2/CARD15, a pathogen receptor detecting bacterial muramyl peptide, as a risk factor of GvHD; however, extensive studies were conflicting (reviewed in Penack et al. 2010).

More recently, the availability of noncultural methods to analyze the whole set of bacteria (called microbiota) has broadened our view as the presence of commensal microbiota and a high diversity of the patients' microbiota associated with substantial protection not only from GvHD but also from systemic and pulmonary infectious complications (Peled J et al, 2016). The exact mechanisms of this protection need still to be defined before translation into new preventive approaches, but beneficial effects of microbial metabolites (such as short-chain fatty acids and indoles) both on epithelial integrity and on immunoregulation have been shown (Swimm et al. 2018; Shono and van den Brink 2018).

43.4 Diagnosis and Scoring

aGvHD is manifested by one or more of the following features: an erythematous skin reaction, cholestatic liver disease, and gastrointestinal dysfunction. The variety of presentations in each organ is provided in more detail in Table 43.2; the syndrome ranges from a mild self-limiting condition to a serious and potentially fatal disorder. Because of the complexity of care of an allo-HCT recipient, it is often very difficult to distinguish the characteristic features of aGvHD from those of other complications such as VOD/SOS, conditioning, and general drug toxicity and infection and consequently to determine the appropriate choice of treatment.

For this reason, it is essential to establish the diagnosis by biopsy of one or more affected organs and confirmation of the characteristic histopathological features (Table 43.3). The targets of the immune response in aGvHD are the epithelial cells including basal and suprabasal cells of the epidermis, the intestinal epithelium, and the biliary duct epithelium, and the characteristic feature is identical, i.e., the presence of infiltrating immune cells close to apoptotic cells known as "satellite cell necrosis."

Table 43.2 Clinical manifestations of acute GvHD

Organ	Clinical manifestation
Skin	Erythematous maculopapular rash, often initially involving the palms and soles May progress to involve the entire body surface and may be pruritic and/or painful In severe cases, bullae may form leading to desquamation
Liver	Cholestasis with or without frank jaundice Cholestatic enzymes comparatively more deranged than transaminases
Gastrointestinal (GI) tract	*Upper*: anorexia, nausea, and vomiting *Lower*: diarrhea, typically green and watery; in severe case diarrhea contains fresh blood and mucosa and is accompanied by abdominal cramps and, on occasion, paralytic ileus

The first classification of aGvHD was developed by Glucksberg et al. (1974). Each organ was staged from 0 to 4 (Table 43.4), and the resultant stages were combined to provide an overall grade (Glucksberg et al. 1974). In 1994 Przepiorka et al. described the outcome of a consensus workshop to develop an improved scoring system that retained most of the characteristics of Glucksberg but dropped the use of the clinical performance score and included upper intestinal symptoms within the definition of aGvHD (Przepiorka et al. 1995). Subsequently, the IBMTR prospectively evaluated a "severity index" against the

Table 43.3 Histopathological findings in acute GvHD

Organ	Histopathological features
Skin	The diagnostic feature is a lichenoid infiltration of the upper dermis and lower epidermis with vacuolation, degeneration, and individual cell necrosis of the cells of the basal layer of the epidermis Grade I: vacuolation of epidermal basal cells Grade II: presence of individually necrotic keratinocytes Grade III: confluent areas of keratinocyte necrosis forming bullae Grade IV: sloughing of the epidermis
Liver	The most consistent histological feature is small bile duct damage, which is usually seen in association with cholestasis and is rare in other complications of HCT The biliary epithelial cells have enlarged hyperchromatic nuclei or small pyknotic nuclei and vacuolated cytoplasm Periportal and midzone hepatocellular necrosis and minimal lymphocytic infiltrates in the portal tract Although there is a histological grading for liver histology, it has no proven prognostic value
Gastrointestinal	"Exploding crypts" within which are necrosis of individual epithelial cells at the periphery of the crypts leaving fragments of nuclear and cytoplasmatic debris Grade I: individual cell necrosis Grade II: loss of individual crypts Grade III: loss of two or more adjacent crypts with ulceration Grade IV: denudation of epithelium

Table 43.4 Staging of acute graft-versus-host disease (according to Harris et al. 2016)

GVHD target organ staging				
Stage	Skin (active erythema only)	Liver (bilirubin)	Upper GI	Lower GI (stool output/day)
0	No active (erythematous) GVHD rash	<2 mg/dL	No or intermittent nausea, vomiting, or anorexia	Adult: <500 mL/day or <3 episodes/day Child: <10 mL/kg/day or <4 episodes/day
1	Maculopapular rash <25% BSA	2–3 mg/dL	Persistent nausea, vomiting, or anorexia	Adult: 500–999 mL/day or 3–4 episodes/day Child: 10–19.9 mL/kg/day or 4–6 episodes/day
2	Maculopapular rash 25–50% BSA	3.1–6 mg/dL	–	Adult: 1000–1500 mL/day or 5–7 episodes/day Child: 20–30 mL/kg/day or 7–10 episodes/day
3	Maculopapular rash >50% BSA	6.1–15 mg/dL	–	Adult: >1500 mL/day or >7 episodes/day Child: >30 mL/kg/day or >10 episodes/day
4	Generalized erythroderma (>50% BSA) plus bullous formation and desquamation >5% BSA	>15 mg/dL	–	Severe abdominal pain with or without ileus, or grossly bloody stool (regardless of stool volume)

Glucksberg criteria but were unable to identify any particular advantage for the new system (Rowlings et al. 1997). In fact, the Glucksberg score was a better predictor of survival and remains in regular use (Cahn et al. 2005). Currently, electronic applications are developed supporting the accuracy of staging and grading of acute GvHD (Schoemans et al. 2018).

Overall clinical grade (based upon most severe target organ involvement):

- Grade 0: No stage 1–4 of any organ
- Grade I: Stage 1–2 skin without liver, upper GI or lower GI involvement
- Grade II: Stage 3 rash and/or stage 1 liver and/or stage 1 upper GI and/or stage 1 lower GI
- Grade III: Stage 2–3 liver and/or stage 2–3 lower GI, with stage 0–3 skin and/or stage 0–1 upper GI
- Grade IV: Stage 4 skin, liver or lower GI involvement, with stage 0–1 upper GI

43.5 Epidemiology

Moderate to severe aGvHD occurs in approximately 40% of all recipients of allo-HCT, but the precise incidence varies considerably depending predominantly on the nature of the donor and the method of GvHD prophylaxis. Without effective prophylaxis, it is an almost inevitable and frequently deleterious complication at least in unrelated matched donor and mismatched family grafts.

43.6 Prevention (also See Chap. 26)

Grade III–IV aGvHD, especially if it turns out to be resistant to first-line treatment, has an extremely poor prognosis despite therapeutic intervention, and consequently, considerable efforts are made to try and prevent its occurrence. The rationale of prophylaxis was originally directed toward prolonged IS of donor T-cell function through the peri- and posttransplant administration of IS agents. Early studies identified the superiority of a combination of the CNI and CSA, with MTX over MTX alone. In practice this combination remains the most frequently used method of prophylaxis although some investigators have replaced CSA with tacrolimus (TAC) since large two phase III randomized studies reported a reduction in the incidence of grade II–IV aGvHD at 32% in recipients of sibling

transplants and 56% in those who received unrelated donor grafts in patients who received TAC plus MTX compared to 44% (sibling) and 74% (unrelated) in those who were randomized to CSA and MTX. However, there was no difference in survival that could be attributed to the nature of the GvHD prevention (Ratanatharathorn et al. 1998; Nash et al. 2000). Recently, investigators have also reported the efficacy of newer agents such as mycophenolate mofetil (MMF) and sirolimus (SIR). Whereas MMF has not been tested in large randomized trials, the combination of TAC and SIR was compared with that of TAC and MTX in a phase III randomized study showing equivalent efficacy but differences in toxicity (Törlén et al. 2016).

An alternative approach to GvHD prophylaxis is to consider removal of donor T cells either ex vivo prior to infusion or in vivo before and/or after infusion using polyclonal (anti-thymocyte globulin [ATG]) or MoAb. A similar effect can also be achieved by positive selection of CD34+ stem cells. These techniques, collectively known as TCD, are extremely efficient in preventing acute and chronic GvHD and were in widespread use in the 1980s and 1990s. Unfortunately, they were rapidly identified as contributing to an increased risk of infection and disease relapse and subsequently became confined to situations in which the risk of GvHD is increased, e.g., recipients of mismatched and haploidentical transplants where the risk of death from GvHD outweighs the risk of later disease recurrence. In unrelated donor HCT, polyclonal ATG has become a major player as two randomized trials showed positive effects mainly on chronic GvHD (Finke et al. 2009; Walker et al. 2016).

Recently, a prospective, double-blind phase III trial to investigate the effect of ATG in the setting of MUD HCT showed that grade II–IV acute GvHD and moderate–severe cGvHD was lower in ATG recipients but the overall survival was lower in ATG recipients (Soiffer et al. 2017). This could be related to higher ATG levels in patients with low lymphocyte counts following TBI which might translate in subsequent infectious complications and EBV-related posttransplant lymphoproliferative disease. Thus, balancing suppression of long-term GvHD versus suppressing anti-infectious defense is an ongoing challenge in GvHD (Gagelmann et al. 2017).

Other studies have explored alternative methods of aGvHD prophylaxis including the infusion of an expanded population of T-regulatory cells at the time of stem cell infusion and partial TCD such as depletion of α/β T cells or elimination of alloreactive T cells after in vitro or in vivo activation. In this context, the administration of PT-CY in order to eliminate early activated donor T cells has gained substantial interest particularly in the context of haploidentical transplantation, and further studies comparing the more complex T-cell depletion approach with the simple approach of PT-CY are currently performed (Kanakry et al. 2016). In a prospective, randomized study on 431 adults given reduced-intensity conditioning and 6/6 related or 7–8/8 unrelated donor grafts, GvHD prophylaxis consisted of either PT-CY, tacrolimus and MMF, or tacrolimus and MTX (Holtan SG Blood 2022; LBA). One-year GvHD relapse-free survival (GRFS) was significantly better in the PT-CY study arm (52.7% vs. 34.9%, $p < 0.001$) due to significantly lower incidences of acute and chronic GvHD.

Recent trials showed the activity of abatacept for aGvHD prevention (Watkins et al. 2021) leading to the FDA approval of abatacept in combination with a CNI and MTX for allo-HCT with a matched or 1-allele mismatched unrelated donor.

Also vedolizumab, an antibody directed against α4/β7 integrin which is selectively expressed in the GI tract, was tested in a prospective phase III trial and showed a significant reduction of GI aGvHD (ChenYB et al, 2019, Floisand Y et al, 2021).

43.7 Treatment

Grade I aGvHD, by definition affecting only the skin, can often be effectively treated with topical steroids alone. Early systemic treatment of grade I GvHD has been tested but showed no long-term advantage. More advanced grades require systemic therapy, and the mainstay of treatment

remains high-dose methylprednisolone (or equivalent), usually at a dose of 2 mg/kg/day, continued for 7–14 days and followed by a gradual reduction in dose (Penack et al. 2020). Patients with mild upper GI GvHD may start on lower doses with concomitant topical treatment; higher doses of steroids resulted in more infectious complications without superior long-term response. The chance of response decreases with increasing grade of GvHD, but in general approximately 40–50% of patients will demonstrate a response. Reductions in steroid doses may be followed by an exacerbation of symptoms that can sometimes be settled by simply increasing the dose and reducing more slowly on the second occasion. Achieving a balance between the levels of IS required to control aGvHD and retaining a degree of immunocompetence against microbial infection is challenging, and viral and fungal infections are frequent complications of prolonged steroid therapy. Anti-infective prophylaxis should be considered for all such patients. Among several candidates for first-line combination treatments, the most promising combination of steroids and MMF has been taken forward to a phase III study against steroids alone but failed to show superiority for the combination again due to an increased rate of infectious complications. Thus, so far no single agent has shown superiority of results when combined with corticosteroids for first-line treatment (Martin et al. 2012; Rashidi et al. 2016).

Failure to respond to standard steroid doses (defined as progression within 3–5 days of starting treatment or an incomplete response by 7–14 days) or recurrence after initial dose reduction (steroid dependence) will necessitate second-line treatment. In this context many agents have been tried alone or in combination with corticosteroid.

A multicenter phase 3 trial reported that ruxolitinib was superior to standard care for grade 2–4 SR-aGVHD (Zeiser et al. 2020). In this trial, 309 patients ≥12 years old were randomly assigned (1:1) to ruxolitinib (10 mg by mouth, twice daily) versus the investigator's choice of therapy; control therapy was chosen by the investigator at the time of randomization from the following: anti-thymocyte globulin, extracorporeal photopheresis, mesenchymal stromal cells, low-dose methotrexate, mycophenolate mofetil, everolimus, sirolimus, etanercept, or infliximab. At day 28, compared with the control group, ruxolitinib achieved superior overall response rate (ORR; 62 versus 39%; odds ratio 2.64 [95% CI 1.65–4.22]) and complete response (CR; 34 versus 19%) (Zeiser et al. 2020). Superiority of ruxolitinib was maintained at day 56 (40 versus 22%) with all grades of disease and affected organs. There is no consensus treatment for SR-aGVHD beyond ruxolitinib. The preferred treatment varies among institutions and is influenced by availability, cost, and patient preference.

Most agents often result in short-term control, but durable effects are relatively infrequent, and the outcome of refractory aGvHD is dismal with approximately 80% mortality, especially if the lower GI tract is involved.

Responses have been reported with extracorporeal photopheresis administered at least twice a week on a weekly basis, and outcome seems to be superior with less toxicities occurring (Jagasia et al. 2013) (see Chap. 66).

In 2006, Ringden et al. reported the successful use of mesenchymal stromal cells (MSC) in a small group of patients with refractory severe aGvHD, and later this group described a response rate of >50% in a larger group of patients (Munneke et al. 2016). MSC exert immunosuppressive effects in a non-HLA-restricted manner and like Tregs offer interesting and novel strategies for the management of this potentially fatal complication although long-term results need to be established in future trials (Le Blanc et al. 2008). Currently, a prospective phase III trial evaluates the efficacy of MSC versus ruxolitinib in patients with SR-aGvHD (IDUNN trial).

The administration of pooled fecal allogeneic microbiotatherapy (FMT) in 81 patients with SR-aGvHD of the GI tract reportedly resulted in an ORR of 56% including 37% CR after a median of three doses and translated to increased OS in responding patients (Malard F et al, 2023). Currently, a phase III prospective study in SR-aGvHD patients is ongoing to evaluate the efficacy of FMT.

While classical IS regimens inhibit a signal pathway or cytokine receptor, novel strategies target CD28 (e.g., Aurora kinase), cell migration (ROCK), or growth factor signaling (e.g., MEK)

(Zeiser et al. 2017). These inhibitors were tested in preclinical studies and showed promising activity (Zeiser R, Blazar B 2017, Hill et al. 2018).

Another novel approach is infusion of alpha-1 anti-trypsin which exerts anti-inflammatory effects and stimulates regulatory T cells. Two recent phase II trials showed CR rates of 35% and OR rates of 60% of the patients on day 28 after treatment starts (Marcondes et al. 2016; Magenau et al. 2018).

43.8 Future Perspectives: Biomarkers, Risk-Adapted Treatment, and Tissue Regeneration

The difficulties to improve results in SR-aGvHD underline that steroid resistance might not just represent resistance of alloreactive T cells but loss of immunoregulation and tissue tolerance (Wu and Reddy 2017) which is difficult to overcome by classical immunosuppressants. Besides new approaches of modulation, potential solutions might be earlier risk adapted or even preemptive treatment strategies which require, however, reliable and reproducible identification of these patients. Recently, clinical risk scores (MacMillan et al. 2015) and novel biomarkers have been reported. The strength of these biomarkers for early identification of high-risk patients at day 7 after HCT or at onset of GvHD has been proven in large multicenter consortia and needs now confirmation by trials on biomarker-guided treatment strategies (Vander Lugt et al. 2013; Hartwell et al. 2017; Levine et al. 2015). The strength of the current biomarkers and scores are partially explained by the fact that they identify GI GvHD as the most severe and deleterious manifestation in an early phase of the disease. Besides the immunosuppressive approaches for aGVHD novel treatment that aim at regenerating the intestinal tract including IL-22 and glucagon-like peptide 2 (GLP-2) have been tested in mouse models of GVHD (Norona et al. 2020; Hanash et al. 2012) and in early clinical trials (Ponce et al. 2023). The data show acceptable toxicity and signs of efficacy that need to be validated in prospective phase III trials.

Key Points

- Acute GvHD occurs until day 100 as classical acute GvHD and beyond day 100 as delayed acute GvHD.
- As treatment options are limited beyond the use of corticosteroids, careful selection of GvHD prevention is essential.
- In the USA abatacept has been approved for GvHD prevention in allo-HCT with unrelated donors.
- Besides classical IS agents like CNI, MTX or MMF, and m-ToR inhibitors, partial TCD (IV serotherapy, depletion of T-cell subpopulations, elimination of alloreactive T cells, or PT-CY) are possible options for prophylaxis.
- Ruxolitinib has received EMA/FDA approval for treatment of SR-aGvHD.
- Institutional standards of treatment and supportive care are essential.

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